Jordan River Water Use Plan

Lower Jordan River Fish Index Study

Reference: JORMON#2

Lower Jordan River Fish Index Study – Year 3 Results (2007)

Study Period: August – September 2007

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December 15, 2008

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Year 3 Results (2007)

Prepared For

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Water Use Planning
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December 15, 2008

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EXECUTIVE SUMMARY

The Lower Jordan River Fish Index Study was instigated to assess the biological benefits of a 0.25 m^3/s fish flow release from Elliott Dam (initiated in January 2008). The study design incorporates 3 years of pre flow release monitoring followed by 3 years of post flow release monitoring. Performance measures include fish abundance, fish condition factor, and continuity of habitat. The geographic scope includes the Jordan River from Elliott Reservoir downstream to the Jordan River Generating Station tailrace. This report presents the results of year 3 of the study (2007). Three main tasks were identified in the project terms of reference for year 3:

1) Conduct a fish sampling program to estimate standing stock and condition factor of rainbow trout at previously established index sites.

2) Complete habitat inventories of subsections of each reach of the lower Jordan River.

3) Continue environmental monitoring at the 5 index sites established in 2005 including servicing of water temperature loggers and collection of water samples for lab analysis (pH, conductivity, alkalinity, total dissolved copper, and total copper).

In 2007, fish population surveys were conducted at the same 17 index sites established in 2005 and 2006. Nine of these sites received 3-pass removals and the other 8 received 1-pass removals. Maximum likelihood estimates were used to compute the fish population within the 3-pass sites. Regression analysis using data from 2005 to 2007 3-pass sites, and from Griffith (1996), was used to derive an expansion factor for estimating the populations within 1-pass sites. All salmonids captured were measured for length and weight in order to derive Fulton's condition factor.

Standing stock was computed in terms of density and biomass of rainbow trout by age group. Geometric means were used to summarize these performance measures. For condition factor, arithmetic mean of all age groups combined was computed. Results for 2007 were as follows:

	Density (fish/100 m^2)	Biomass (grams/100 m^2)	Condition Factor (K)
0^+ Fry:	6.4	15	1.08
1^+ Parr:	0.7	4	

Habitat inventories were conducted on 6 sections of the lower Jordan River ranging in length from 171 to 482 m for a total survey length of 1,971 m. Connectivity of habitat was poorest in the 2 reaches immediately downstream of Elliott Dam where 70% (306 m) of Reach 7 was dry channel and 12% (145 m) of Reach 6 b was dry channel. Below this, physical connectivity improved, however fish movement was still frequently impeded by short sections of very shallow water at hydraulic controls. Another finding was that riffle habitats (the main source of insect production in streams) tended to be a minor proportion of available habitat types (11% based on stream length, 6% based on stream wetted area).

Environmental monitoring involved the collection of water quality samples at 5 of the fish population index sites, and monitoring of water temperature loggers at 4 sites. As in previous years, water quality results showed that copper concentrations increased by an order of magnitude downstream of the mine tailings (Reach 2); however, 2007 levels in this reach were about half of those found in 2006. This was also the first year in which juvenile salmonids were captured in the copper affected reach, though these fish exhibited extremely poor condition factor (0.90). Thus, it appears that in 2007 copper concentrations in Reach 2 had declined below the threshold that caused juvenile rainbow to avoid this section of water, but still adversely affected their growth.

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ACKNOWLEDGEMENTS

A number of BC Hydro staff contributed to the success of year 3 of the Jordan River Fish Index Study: Ian Dodd provided capable handling of the administrative and coordination tasks, and assisted in the field; Duane Walsh and Brent Mockford provided support with lock-out safety procedures; Ian Dodd and Alf Leake reviewed the manuscript and provided helpful suggestions for report improvement. I am especially grateful to the Pacheedaht First Nation for providing Jeff Jones as my field assistant. The Jordan is an extremely difficult and strenuous river to work in, and Jeff's endurance and field capabilities were a major asset to the field program. Thanks also to Helen Dunn for coordinating activities on behalf of the Pacheedaht First Nation. Lastly, appreciation is extended to Lew Carswell for supplying his expertise in the aging of scale samples.

1. INTRODUCTION

The Jordan River Water Use Plan (JOR WUP) was initiated in April 2000, submitted to the Comptroller of Water Rights (CWR) in 2002, and approved by the CWR on July 20, 2004 (BC Hydro 2005b, Attachment A). The terms of the Jordan WUP included instigation of a fish flow release of 0.25 m^3/s from Elliott Dam in 2007, and implementation of various monitoring projects to assess fish habitat, abundance, and physical conditions in the river before and after the flow release. This report presents the results for 2007 for one of these monitoring projects, the Fish Index Monitoring Study, and represents year 3 of this study. Due to various construction setbacks, installation of the flow release structure at Elliott Dam was not completed until January 2008 with instigation of the flow release on January 17, 2008. Thus, the 2007 results represent the third year of before-treatment data. After-treatment data will be provided by monitoring in the ensuing 3 years.

The goal of the Lower Jordan River Fish Index Study is to assess the biological benefits of the fish flow release using fish abundance, fish condition factor, and continuity of habitat as performance measures (BC Hydro 2005a, Attachment B). The geographic scope of the project includes mainstem reaches of the Jordan River from Elliott Reservoir downstream to the Jordan River Generating Station tailrace. Three tasks were identified in the project terms of reference for year 3:

1) Conduct a fish sampling program to estimate standing stock and condition factor of rainbow trout at index sites in the lower Jordan River

2) Complete habitat inventories on index subsections of each reach of the lower Jordan River (encompassing fish population sites).

3) Continue environmental monitoring at the 5 index sites established in the lower Jordan River in 2005. This will include servicing of water temperature loggers and collection of water samples for lab analysis of water quality (pH, conductivity, alkalinity, total dissolved copper, and total copper).

The purpose of this report is to present the results for year 3 (2007) of this study. Results for years 1 and 2 were presented in Burt (2006) and Burt (2007), respectively. Assessment of the benefits of the fish flow release will commence in year 4 of the program.

2. BACKGROUND

Previous Work

Previous fisheries studies on the Jordan River include a biophysical and fish production assessment by Griffith (1996), an instream flow study by Cascadia Biological Services (2001), and a pink salmon incubation study by Wright and Guimond (2003). A synopsis of these studies was provided in the year 1 report (Burt 2006).

Study Area

The Jordan River is located on the southwest coast of Vancouver Island 72 km by road from the city of Victoria. The river originates in the Seymour Mountain Range in south central Vancouver Island and flows in a south-westerly direction before emptying into Juan De Fuca Strait adjacent to the community of Jordan River. It has a drainage area of 184 km^2 and a mean annual discharge at the mouth of approximately 13.7 m^3/s (Cascadia 2001).

In terms of the Provincial Ecoregion Classification System, the Jordan Watershed spans two distinct ecosections. The lower Jordan River and western tributaries of the upper river lie in the Windward Island Mountains (WIM) Ecosection (Coast and Mountain Ecoprovince), while the mainstem of the upper river and its eastern tributaries lie within the Leeward Island Mountains (LIM) Ecosection (Georgia Depression Ecoprovince). Climate in the WIM Ecosection is dominated by the arrival of frontal systems from the Pacific Ocean resulting in heavy rainfall during the winter months. In contrast, the LIM Ecosection occurs on the leeward side of the Vancouver Island Mountains and experiences a rainshadow effect resulting in much lower precipitation than the NIM Ecosection. Details of this classification system and associated characteristics are described in Demarchi (1996).

The Jordan River hydroelectric project was completed in 1911 and rebuilt in 1971. Current facilities include three dams, two reservoirs, a headpond, a tunnel/penstock water delivery system, and a 175 MW powerhouse on the lower Jordan River (BC Hydro 2003). The dams include Bear Creek Dam and Jordan Diversion Dam, which impound Bear Creek Reservoir (7.5 km^2) and Diversion Reservoir (18 km^2), and Elliott Dam which impounds Elliott Headpond (1.6 km^2). The powerhouse is located on the west side of the lower Jordan River 900 m above the mouth. This powerhouse replaced an older facility and tailrace in 1972 (located on the east side of the river).

The study area for the fish index monitoring program encompasses the mainstem from Elliott Dam to the Jordan River Generating Station tailrace, a distance of 7.8 km (Figure 1). Drainage area of the study region is 37 km^2, which represents 20% of the total Jordan Watershed (Cascadia 2001). Under current operation of Elliott Dam, no water is released into the lower Jordan mainstem except when spilling is required. Thus, flows in the study reaches are generally dependent on input from local tributaries. The most significant of these include Sinn Fein, Winkler, and Nuala Creeks, which drain the west side of the river, and one unnamed creek just above Nuala that drains the east side of the river. As indicated in the Jordan WUP, this situation will change in year 3 of the study when BC Hydro commences release of a fisheries flow of 0.25 m^3/s from Elliott Dam.

Physical characteristics of the study portion of the Jordan River include a channel that varies from confined to entrenched within steep valley walls. The overall map gradient is about 4%, but individual reaches range from 1.5% to 10% (Cascadia 2001). Channel types vary from riffle-pool sequences in lower gradient sections, to cascade-pool sequences in steeper sections. Substrates tend to be dominated by large boulders with cobbles and/or bedrock as subdominant categories. Spawning gravels are scarce but do occur in isolated pockets.

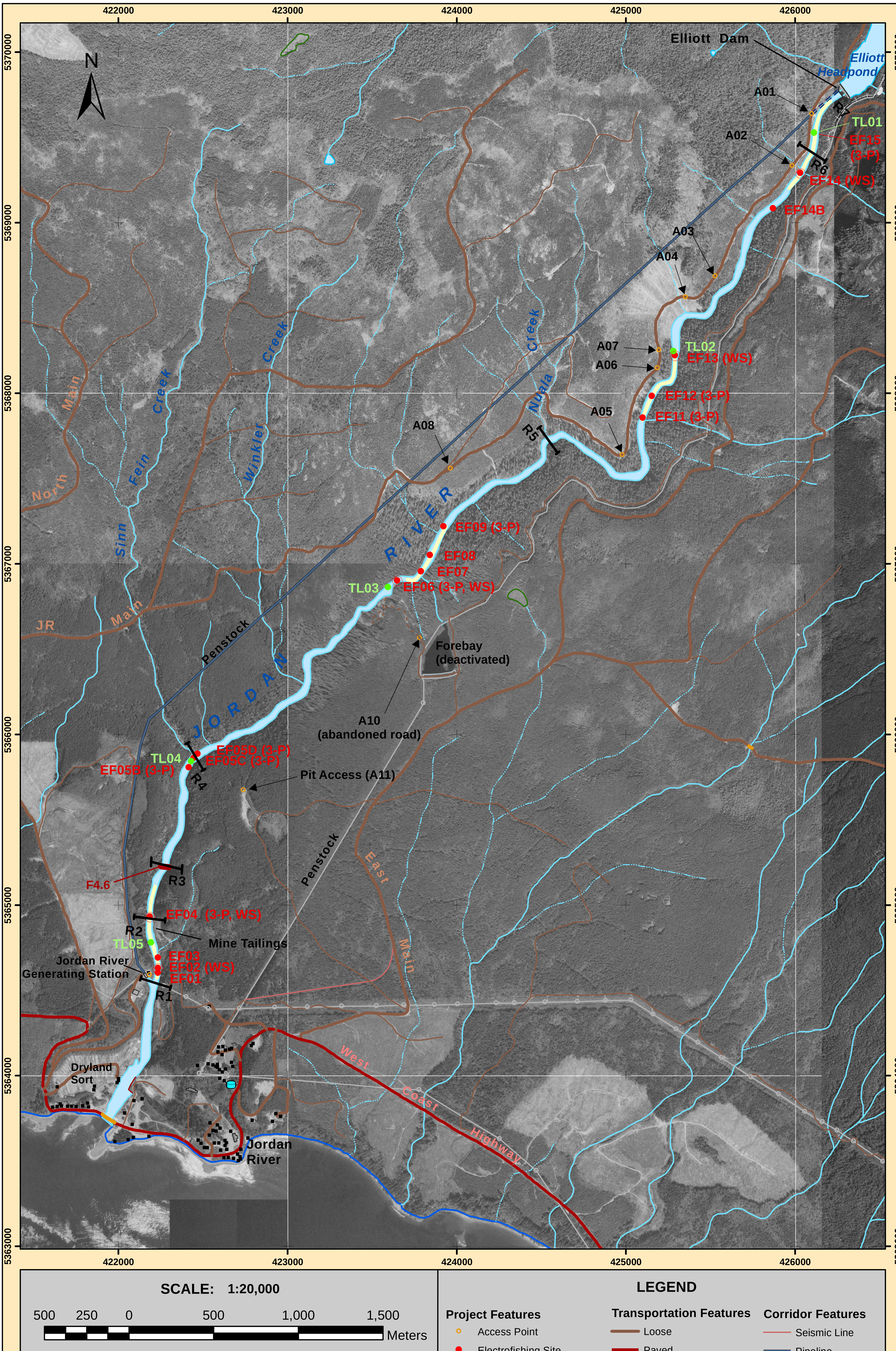

Figure 1. Map of the lower Jordan River showing electrofishing, temperature logger, and water sample sites, habitat survey sections, and access points.

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3. METHODS

3.1 Fish Sampling

Field Procedures

In 2007, fish sampling was conducted at the same 17 sites established in 2005/2006. Locations of these sites are shown in Figure 1. Sample timing was September 4–8, 2007. Flows were at summer base levels, however, 2007 was a wetter summer than in 2005/2006 and flows were somewhat higher than previous sample years.

The approach used to sample fish abundance involved either 3-pass or 1-pass electrofishing removal techniques. Nine of the 17 sites received 3-pass removals while the remaining 8 sites received 1-pass removals (3-pass sites are designated with “3-P” adjacent to their name in Figure 1). Due to extremely low flows in the river, aquatic habitat was primarily glides with little or no surface flow on connecting riffles. Thus, for most sites stop nets were not needed to enclose discrete sample areas. The electrofishing was performed by either a two or a three person crew using a Smith-Root Model 12A backpack electrofisher. Each “pass” generally involved a circuit upstream through the site and then back down through the site. After each pass, fish were identified, measured for fork length (to nearest 1.0 mm), and weighed (nearest 0.1 g, OHAUS portable electronic balance, Model C-505). Alka-seltzer tablets (CO2) were used as an anaesthetic. Scales were collected on a subsample of fish for later age determination. Details of these sample procedures are described in Anon. (1995).

Upon completion of the fish sampling, a transect was set across the site at a representative hydraulic location using a 30 m tape. Depth and velocity data were recorded at 20–50 cm intervals along the transect. These data were later used to determine the amount of suitable rearing habitat within the site, and provided a means of prorating fish densities by the amount of suitable habitat. Depths were measured using a 1.5 m top-setting rod and velocities were taken at 40% of the depth (from the bottom) using a Swoffer flow meter (Model 2100) mounted to the rod.

Analysis

Analysis of electrofishing data first involved assignment of age to all captured fish so that population estimates could be generated by age group. Age designation was determined using length-frequency analysis in conjunction with scale age results. Scale aging involved placing each scale sample under a microscope and taking a digital photograph of the best scale. The resultant images were then used for assessing age. Scale aging was performed by Lew Carswell, retired fisheries technician from the Ministry of Environment.

Methods used to compute population size (by age group) differed between 3-pass and 1-pass sites. For 3-pass sites, Bayes’ maximum likelihood estimates (MLE) were generated using a computer program developed by BC Hydro ("PopEst7.xls", Bruce and Z’Graggen 1995). The program also generated variances, standard errors, and 95% confidence limits for each estimate.

Population estimates were converted to fish density (fish/100 m^2) by dividing the estimate by the site area and multiplying by 100.

For single pass sites, population estimates were computed using an expansion factor based on data from multiple removal sites. The expansion factor was derived from linear regression of maximum likelihood population estimates (dependent variable) on the number of fish caught in removal 1 (independent variable) (Kruse et al. 1998, see Lobon-Cervia and Utrilla 1993, Meyer and Lamansky 2002). Data used in the regression included 3-pass sites from 2005, 2006 and 2007, as well as 4 2-pass sites completed in 1994 by Griffith (1996). Because numbers of fish captured at a given site were relatively low, the analysis was run with all age groups combined. I also set the intercept to zero. Figure 2 shows the results of this analysis. Four sites were identified as outliers and were removed from the regression (EF06 and EF11 in 2006; EF04 and EF12 in 2005). The x coefficient (1.262) was used as the expansion factor to derive population estimates for 2007 single pass sites. Confidence limits for these population estimates were based on the 95% prediction intervals (outer lines in Figure 2).

Figure 2. Relationship between number of rainbow trout captured by removal 1 and the corresponding population estimate (all age groups combined) based on electrofishing in the study area during 2005 – 2007 (this study) and in 1994 (Griffith 1996). The outer lines depict the 95% prediction intervals.

Fulton condition factor was calculated for all captured salmonids using the following equation (Anderson and Neumann 1996):

$$K = \left(\frac{W}{L^{3}_{FL}} \right) \times 10^{5}$$

where: K = *Fulton condition factor*

W = *weight in grams*

L_{FL} = *Fork length in millimetres*

3.2 Habitat Inventory

The goal of the habitat inventory was to collect quantitative and qualitative information on the nature of fish habitat in the study area. This will provide a means to determine changes in fish habitat after initiation of the fish flow release (January 2008). A total of 6 sections were inventoried during August 30, 31 and September 9, 2007. Survey sections were selected so as to encompass the 2007 electrofishing sites. Stream lengths surveyed ranged from 171 m to 482 m. Locations of survey sections are shown in Figure 1.

Habitat surveys followed methods outlined in Johnston and Slaney (1996). This involved walking the stream thalweg with a running hip chain and recording distance of the top and bottom of each habitat unit encountered. GPS coordinates were collected at the origin of each survey section in order to establish its location on digital TRIM maps. All habitat units within a given survey section were assessed and information recorded on prepared field forms. Data collected for each habitat unit included habitat unit number, type (pool, glide, riffle, or cascade), length (m), mean depth (m, average of 3 measurements), maximum depth (m), residual pool depth (m, pools and glides only), mean wetted width (m, average of 3 measurements), average velocity (m/s), substrate composition (percent bedrock, boulder, cobble, gravel, and fines), and dominant/subdominant cover types and their relative percentages. Average velocity of habitat units was determined from the equation:

$$v_{ave} = \frac{Q}{d_{ave} \times w_{ave}}$$

where: V_{ave} = *average velocity in m/s*

Q = *discharge in* m^3/s

d_{ave} = *average depth in m*

w_{ave} = *average width in m*

Discharges for determination of v_{ave} were obtained from the Inflow Monitoring Program (Burt and Hudson 2008).

Habitat units situated on electrofishing sites received additional data collection including more detailed information on cover (percentage of all categories) and substrate (D_{90}, D_{max}, compaction, and degree if infilling).

3.3 Environmental Monitoring

Environmental monitoring involved 1) collection of temperature, pH, and conductivity at all electrofishing sites with handheld meters, 2) collection of water samples at 5 sites, and 3) reinstallation of continuous water temperature data loggers at 4 of the 5 sites established in 2005. Reinstallation of the water temperature loggers was necessary as all loggers were washed away by the November 2006 flood. Locations, dates, and water quality parameters assessed are listed in Table 1. Locations of all sites are provided in Figure 1.

Table 1. Summary of Water quality monitoring conducted during the 2007 Fish Index Study.

Parameters	Sample Method	Locations	Dates
Water temperature, pH, conductivity	Handheld Meters	All 17 electrofishing sites	Sept. 4–8, 2007
pH, conductivity, alkalinity, total copper, total dissolved copper	Water Samples (WS)	EF02, EF04, EF06, EF13, EF14	Sept. 9, 2008
Continuous temperature monitoring	Temperature logger (TL) set to record temperature at 15 minute intervals	TL01 to TL05 in Figure 1	Reinstallation Dates: TL01 - July 4, 2007 TL02 - Aug. 30, 2007 TL03 - July 5, 2007 TL04 – not installed TL05 – Sept. 8, 2007

Handheld meters used at electrofishing sites included a Taylor digital thermometer (model 9847) for temperature, an Oakton waterproof pHTestr 10 for pH, and a Horiba Twin B-173 for conductivity. The water samples were collected in 500 ml plastic bottles (3 replicates per site), stored on ice in a cooler, and delivered to North Island Labs (Courtenay) approximately 24 hrs after collection. The lab was instructed to analyze two replicates from each site with the third retained for analysis in the event of high variance among results. Continuous water temperature monitoring used Stowaway temperature tidbits (Onset Computer Corp.). These were mounted inside 18 cm lengths of 3.8 cm (1½ inch) PVC pipe weighted with 4.5 kg (10 lb) cannon balls. Each apparatus was tethered to an anchor pin in a nearby rock with 9.5 mm (3/8 inch) steel core rope.

4. RESULTS

4.1 Fish Abundance and Condition Factor

Summary of 2007 Electrofishing Sites

In 2007, fish population sampling was conducted at 17 sites (9 3-pass sites, and 8 1-pass sites). Site locations are shown in Figure 1, while a summary of each site's characteristics and capture statistics can be found in Appendix B. The distribution of sites per reach and number of each fish species captured per site are given in Table 2. A total of 140 rainbow trout, and 3 sculpins were captured in 2007. The "rainbow" at EF01 through EF04 may actually have been steelhead given the anadromous access to this site. The sculpins caught at EF02 and EF03 were probably coastrange sculpins (*Cottus aleuticus*) as the first dorsal fin did not display a distinct black spot (McPhail and Carveth 1999).

Table 2. Distribution of electrofishing sites by reach, and number of passes and total catch for each site completed on the lower Jordan River in September 2007.

Reach	Reach Type	Number of Sites	Sites	No. of Passes	Total Catch By Species
R2	Anadromous	3	EF01 EF02 EF03	1-pass 1-pass 1-pass	RB/ST - 3 RB/ST – 1 CC - 1 RB/ST – 17
R3	Anadromous	1	EF04	3-pass	RB/ST – 51 CC – 2
R4	Resident	3	EF05B EF05C EF05D	3-pass 3-pass 3-pass	RB – 4 RB – 2 RB – 6
R5	Resident	4	EF06 EF07 EF08 EF09	3-pass 1-pass 1-pass 3-pass	RB – 19 RB – 2 RB – 8 RB – 6
R6	Resident	5	EF11 EF12 EF13 EF14 EF14B	3-pass 3-pass 1-pass 1-pass 1-pass	RB – 11 RB – 6 RB – 4 No fish No fish
R7	Resident	1	EF15	3-pass	No fish
Totals		17			RB/ST – 140 CC – 3

Notes: RB = rainbow trout, ST = steelhead trout, CC = sculpins

The presence of salmonids at sites EF01, EF02, and EF03 was a first for these sites. Sampling at these sites in 2005, 2006, and by Griffith in 1994 (Griffith 1996) never encountered salmonids, and it was believed that elevated copper concentrations from nearby mine tailings caused juveniles to avoid this section of water. Possible explanations for the presence of salmonids at these sites in 2007 may be the higher flows experienced in summer 2007 (dilution factor), and/or that the November 2006 flood scoured out some of the copper source material. The scour effect of the flood were apparent throughout the river in that anywhere there was soft bank material, the banks were truncated to a height of 2-3 m (including the mine tailings bank).

Density and Biomass

Standing stock of rainbow trout in the Jordan River was assessed in terms of density and biomass. To derive density, maximum likelihood population estimates per age group (3-pass sites), and predicted population estimates (1-pass sites), were converted to fish density (fish per 100 m^2) based on site areas. Density values were not adjusted by habitat suitability criteria (from the depth/velocity transect data) because it was found that such adjustments produced unreasonable expansion of observed densities. Thus, results presented are restricted to observed densities.

Figure 3 summarizes results of the density calculations. Detailed capture statistics can be found in Appendix A. As in previous years, 2007 fry densities were highly variable, ranging from 0 to 52 fry/100 m^2. Field observations suggested that the presence of fry was related to the occurrence of appropriately sized spawning gravel within the sampled habitat unit or in close proximity to it. Densities of 1^+ parr were also variable with a range of 0 to 6 fish/100 m^2. Age groups older than 1^+ were not identified by the length-frequency analysis, however, scale aging indicated that some of the fish designated as 1^+ may actually have been 2^+ fish. Geometric mean density in 2007 was 6.4 FPU for fry (95% CL 2.5 – 14.6) and 0.7 FPU for 1^+ parr (95% CL 0.1 – 1.5). These means were substantially less than the predicted maximum densities of 70 FPU for fry and 11 FPU for 1^+ parr (note: predicted maximum densities were derived by dividing predicted maximum biomass by the 2005-2007 mean weights for fry and 1^+ parr; predicted maximum biomass is discussed below).

Calculation of fish biomass at Jordan River fish index sites was achieved by multiplying density per age group at each site by mean weight per age group at each site (g/100 m^2). Figure 4 shows these results (see Appendix A for details). For sites where fry were caught, biomass ranged from 1 to 164 g/100 m^2. For sites where 1^+ parr were caught, biomass ranged from 20 to 152 g/100 m^2. Geometric mean biomass based on all sites was 15.4 BPU for fry and 3.5 BPU for parr.

The predicted maximum biomass in Figure 4 (241 g/100 m^2) was calculated from a recent model[1] by Ron Ptolemy (MWLAP, Victoria) based on alkalinity and stream enrichment state. For the Jordan River, a non-enrichment state was assumed (a yes/no parameter in the model) and an alkalinity of 18 mg/L was used (2005 – 2006 mean from samples collected by this study). Biomass

[1] Maximum Biomass = $0.65 \times 65.6 \times (ALK)^{0.6}$ (Ron Ptolemy, MWLAP, Victoria, pers. comm.).

levels at the fish index sites were below the predicted maximum which suggests that standing stock was below expectations in 2007.

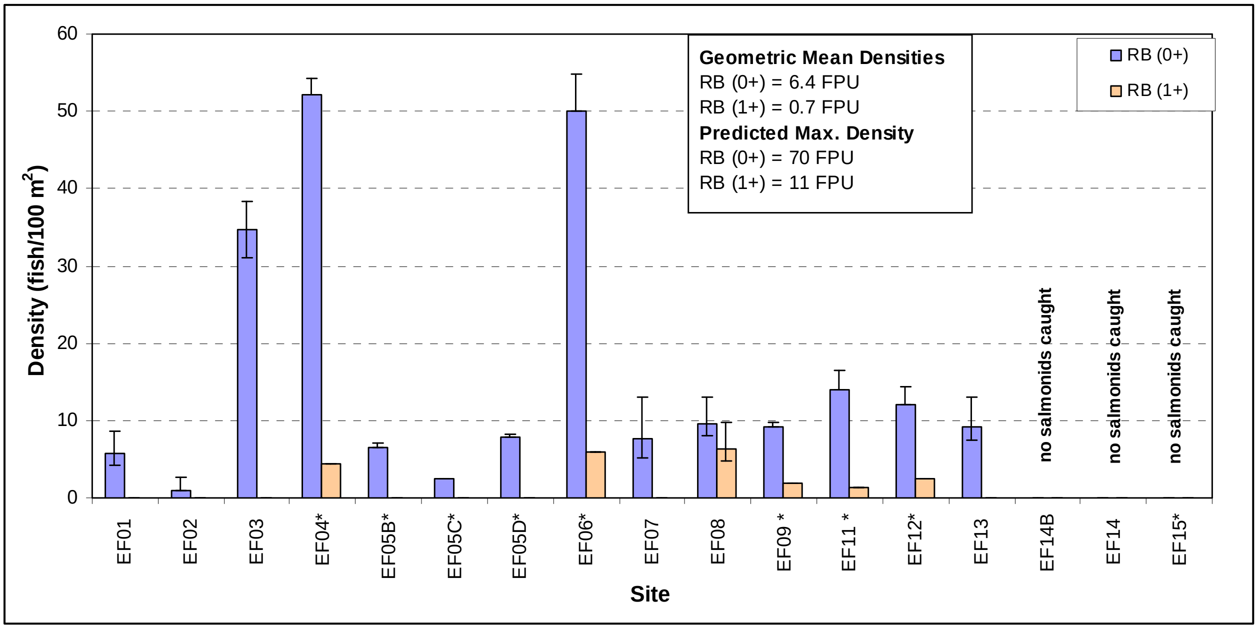

Figure 3. Density of rainbow trout (fish/100 m^2) at Jordan River index sites from September 2007 electrofishing. * indicate sites that received 3-pass removal; all others received 1-pass removal. Error bars denote 95% confidence limits. Some 3-pass sites have no error bars due to 100% catch efficiency (i.e., all fish were caught in pass 1). Geometric means are based on all sites.

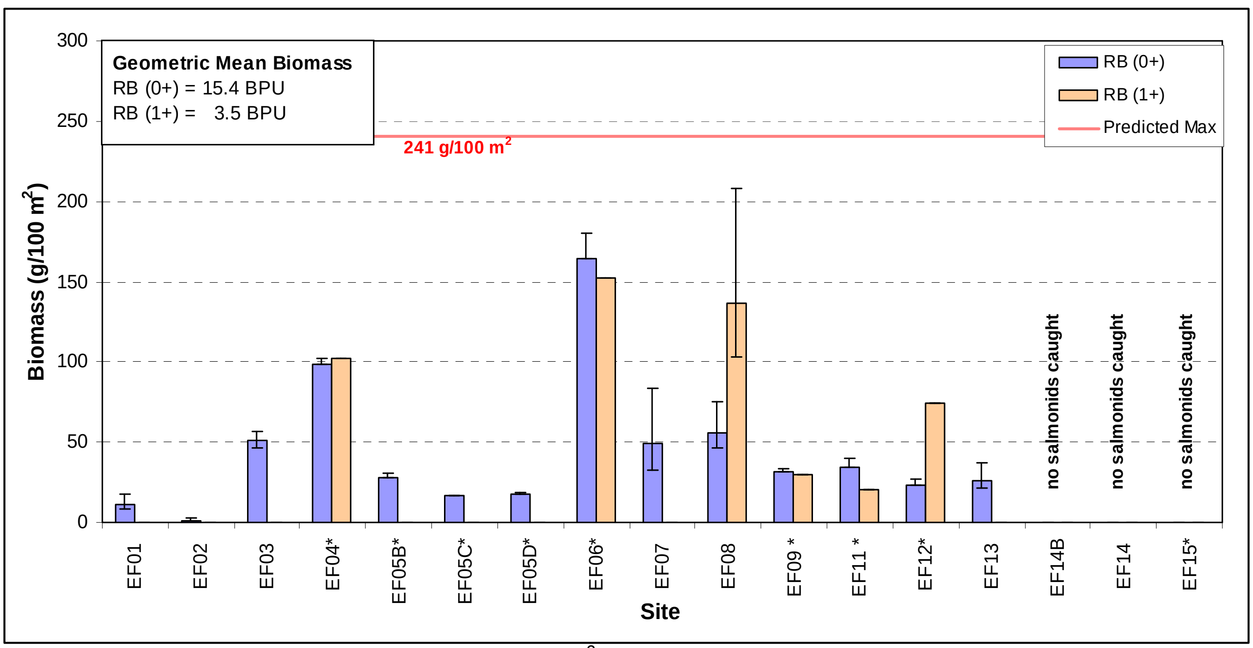

Figure 4. Biomass of rainbow trout (g/100 m^2) at Jordan River index sites from September 2007 electrofishing. The predicted maximum (241 g/100 m^2) is based on a recent model by Ptolemy (pers. comm., see text) using an average alkalinity from 2005 and 2006 sampling (18 mg/L). * indicates 3-pass removal sites; all others were 1-pass sites. Error bars denote 95% confidence limits. Some 3-pass sites have no error bars due to 100% catch efficiency.

Length, Weight, and Condition Factor

The relationship between weight and length for rainbow trout captured in 2007 is shown in Figure 5. For comparative purposes, data from 2005 and 2006 are included. Data were fitted in Excel using a power curve of the form *weight* = *a* × *(length)*^{b}. The fitted curves for 2005 to 2007 are very similar suggesting the relationship between length and weight has remained consistent in the pre flow release study years. The power variable (*b*) has remained close to 3 suggesting isometric growth for these years (i.e., shape does not change as fish grow; Anderson and Neumann 1996).

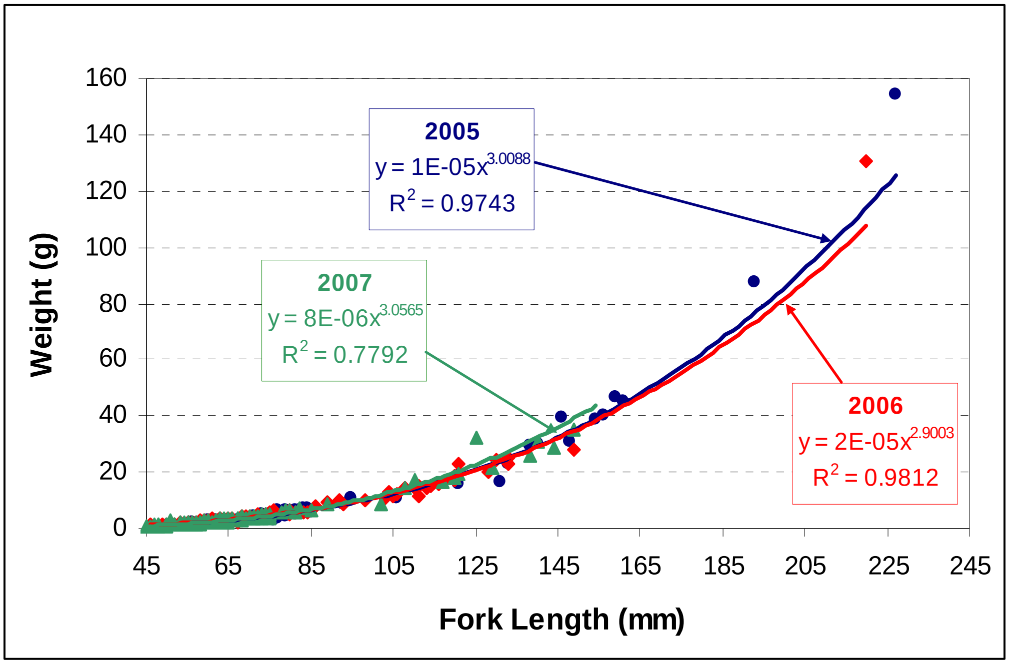

Figure 5. Weight-length relationship for rainbow trout captured by electrofishing on the Jordan River in 2005 (n = 69), 2006 (n = 81), and 2007 (n = 139; 1 outlier removed). Data were fitted with a power curve using Excel.

Table 3 summarizes statistics for length, weight, and condition factor for fish captured in 2007. Overall, rainbow fry captured on the Jordan River in 2007 were smaller (length and weight) than in 2005 and 2006, probably due to cooler growing season than the previous 2 years. Rainbow parr were larger than in 2006 and smaller than in 2005; however, parr comparisons should be treated with caution due to small sample sizes. Overall mean condition factor (all age groups combined) was 1.08 ± 0.03 (95% confidence limits) which was similar to 2005 (1.07 ± 0.04) but less than 2006 (1.13 ± 0.03). As a comparative note, the Provincial default condition factor is 1.05 (Ron Ptolemy, MWLAP, Victoria, pers. comm.).

Table 3. Length (mm), weight (g), and condition factor statistics for fish captured by electrofishing in the Jordan River during September 4–8, 2007

	Age	Statistic					
		n	Min	Max	Mean	SD	95% CL
Fork Length (mm)	0+	128	41	89	60	10	2
	1+	12	102	149	125	15	9
	2+	0					
	3+	0					
Weight (g)	0+	128	0.7	8.8	2.5	1.5	0.3
	1+	12	8.9	35.1	22.4	8.1	4.6
	2+	0					
	3+	0					
Condition Factor	0+	128	0.66	1.96	1.07	0.17	0.03
	1+	12	0.84	1.65	1.10	0.21	0.12
	2+	0					
	3+	0					
	All	140	0.66	1.96	1.08	0.18	0.03

4.2 Habitat Survey Results

Habitat inventories were completed on 6 sections of the Jordan River located within Reaches 2, 3, 5, 6, and 7. Locations of these sections are shown in Figure 1. Distances surveyed per section ranged from 171 m to 482 m, for a combined survey length of 1,971 m. These data were used to estimate habitat unit frequencies and hydraulic characteristics at the reach level (Table 4). Glides and pools were the dominant habitat types in most reaches in terms of both stream length and wetted area. Riffles, the primary areas of aquatic invertebrate production, were relatively scarce (6.2% of overall wetted area for surveyed reaches). These proportions were primarily due to the extremely low flows at the time of the survey associated with the tendency of existing water to go subsurface at riffle habitats (note: flows were estimated at 11 L/s in Reach 2, which equates to ~ 0.07% MAD). The overall percentage of subsurface flow (SS) was 7%, which is about half of the 2006 amount. Reaches with the greatest percentage of subsurface flow were those closest to the dam (R7 –70%, R6b – 12%). Velocities were so low that values were converted from m/sec to cm/sec in order to reduce the number of decimal places in the results. Total wetted area for all surveyed reaches combined was 68,102 m^2 (note: Reach 4 was not surveyed).

Table 4. Summary of habitat unit frequencies and hydraulic characteristics by reach based on stream sections surveyed in the Jordan River during August 29, 30, and September 9, 2007.

Survey Section & Length (m)	Date	Reach No. and Length (m)	Bankfull Width (m)	Habitat Unit Type	Unit Length By Reach (m)	Unit % By Length	Unit Wetted Area By Reach (m^2)	Unit % By Wetted Area	Mean Wetted Width (m)	Mean Velocity (cm/s)	Mean Depth (m)	Mean Max Depth (m)
JOR00875	09/09/2007	R2	39	C	8.2	2.2%	21	0.4%	2.60	10.17	0.14	0.28
337.7		368		G	170.0	46.2%	1,989	41.6%	11.70	2.74	0.27	0.51
				P	142.9	38.8%	2,491	52.2%	17.44	0.26	1.00	1.40
				R	47.0	12.8%	275	5.7%	5.85	7.38	0.14	0.21
				All	368.0		4,775		12.98	0.51	0.53	0.81
JOR01243	30/08/2007	R3	37	C	16.0	7.7%	16	0.5%	1.00	5.79	0.19	0.40
171		342		G	116.2	34.0%	509	16.7%	4.38	0.64	0.46	0.77
				P	167.0	48.8%	2,464	81.0%	14.75	0.21	1.38	2.04
				R	30.0	8.8%	52	1.7%	1.72	4.68	0.28	0.56
				SS	12.8	3.7%	0	0.0%	0.00	0.00	0.00	0.00
				All	342.0	100.0%	3,040		8.89	0.14	0.86	1.32
JOR03956	09/09/2007	R5	29	G	2,210.5	73.2%	21,560	75.3%	9.75	1.20	0.24	0.53
410		3,021		P	262.3	8.7%	3,353	11.7%	12.78	0.98	0.61	1.25
				R	548.2	18.1%	3,734	13.0%	6.81	2.30	0.16	0.28
				All	3,021.0		28,647		9.48	0.80	0.26	0.55
JOR06494	30/08/2007	R6a	31	C	10.3	0.6%	4	0.0%	0.40	4.38	0.12	0.18
481.7		1,709		G	443.1	25.9%	1,981	11.9%	4.47	0.24	0.38	0.63
				P	1,082.1	63.3%	14,567	87.4%	13.46	0.04	0.90	1.39
				R	139.4	8.2%	111	0.7%	0.79	1.54	0.18	0.30
				SS	34.1	2.0%	0	0.0%	0.00	0.00	0.00	0.00
				All	1,709.0		16,233		9.75	0.19	0.11	0.19

Survey Section & Length (m)	Date	Reach No. and Length (m)	Bankfull Width (m)	Habitat Unit Type	Unit Length By Reach (m)	Unit % By Length	Unit Wetted Area By Reach (m^2)	Unit % By Wetted Area	Mean Wetted Width (m)	Mean Velocity (cm/s)	Mean Depth (m)	Mean Max Depth (m)
JOR07857	29/08/2007	R6b	30	G	206.5	16.7%	817	5.6%	3.95	0.24	0.24	0.39
254.0		1,240		P	845.5	68.2%	13,692	94.1%	16.19	0.04	0.87	1.41
				R	43.4	3.5%	36	0.2%	0.82	1.10	0.17	0.33
				SS	144.5	11.7%	0	0.0%	0.00	0.00	0.00	0.00
				All	1,240.0		14,544		11.73	0.02	0.64	1.04
JOR08238	29/08/2007	R7	30	G	96.9	22.0%	312	72.2%	3.22	2.51	0.29	0.44
316.5		440		P	35.3	8.0%	118	27.3%	3.34	0.10	0.48	0.66
				R	1.4	0.3%	2	0.5%	1.70	0.27	0.30	0.00
				SS	306.4	69.6%	0	0.0%	0.00	0.00	0.00	0.00
				All	440.0		432		0.98	1.38	0.10	0.15
All		R2, 3, 5, 6, 7		C	34.5	0.5%	41	0.1%	1.20			
1,970.9		7,120		G	3,243.2	45.6%	27,168	39.9%	8.38			
				P	2,535.1	35.6%	36,684	53.9%	14.47			
				R	809.4	11.4%	4,209	6.2%	5.20			
				SS	497.8	7.0%	0	0.0%	0.00			
				All	7,120.0		68,102		9.56			

Notes:

1. Habitat unit type abbreviations include P = pool, G = glide, R = riffle, C = cascaded, F = falls, and SS = subsurface flow.
2. Survey section names are based on distance in metres from the river mouth to the downstream end of the section; e.g., JOR00875 begins 875 m from the mouth. Distances were calculated by digitizing on TRIM maps in ArcView.

Substrate and cover characteristics of surveyed sections are summarized in Table 5. Boulders were the dominant substrate in all sections while the subdominant substrate was generally gravel or cobble, or a combination of both. An exception was JOR07857 (R6b) where bedrock was the subdominant category. Boulders were also the dominant cover category, and for many habitat units examined, the only cover type. Other cover types present in small quantities were overstream vegetation and deep pools.

Table 5. Summary of mean substrate and cover features for each stream section surveyed during August 29, 30, and September 9, 2007.

Survey Section	Date	Reach	Mean percent of Each Substrate Category					% Boulder Cover
			Bedrock	Boulder	Cobble	Gravel	Fines	
JOR00875	09/09/2007	R2	0.2	36	26	30	8	39
JOR01243	30/08/2007	R3	8	49	9	31	3	42
JOR03956	09/09/2007	R5	0.2	34	29	31	6	32
JOR06494	30/08/2007	R6a	14	50	12	21	3	41
JOR07857	29/08/2007	R6b	35	37	17	9	2	24
JOR08238	29/08/2007	R7	16	58	17	8	1	49

Note: Boulder was the dominant cover type throughout all habitat units surveyed.

4.3 Environmental Monitoring Results

Water samples were collected at five of the electrofishing sites on September 9, 2007 (sites EF02, EF04, EF06, EF13, and EF14; see Figure 1 for locations). Results of laboratory analysis on these samples are provided in Table 6. With the exception of site EF13, alkalinity was higher at upstream locations. This may be related to differences in geographic features between the upper and lower regions of the study area (the upper portion is influenced by conditions of the LIM Ecosection while the lower is dominated by the WIM Ecosection; see Section 2). There is no obvious explanation for the relatively low alkalinity at EF13. Conductivity was highest at EF02, possibly due to salt influence from high tides. Conductivity readings from the handheld meter did not always match those from the laboratory, but in general were within 18% or better of the lab readings. The handheld pH meter gave erratic readings and probably reflects a need to replace the sensor.

The first two water samples were located such that EF02 was downstream of the mine tailings, while EF04 was immediately upstream of seepage sources from the tailings. The copper results in Table 6 show an order of magnitude increase in both dissolved and total copper in waters below the tailings; however, it is noteworthy that concentrations in the mine tailings reach were not as high as in 2005 (DC=0.060, TC=0.064) and 2006 (DC=0.085, TC=0.086). Possible explanations for the lower copper concentrations in 2007 include higher flows at the time the water samples were taken (dilution factor), and/or the scour removal of some of the source material by the November 2006 flood. The fact that rainbow were captured at sites within the copper zone in 2007 (a first for this area) suggests that concentrations were low enough that fish no longer avoided this section. However, it should be noted that the rainbow caught within this zone had very poor condition factors suggesting a physiological effect of copper on these fish or on their food supply (i.e. diminished production of aquatic invertebrates).

Table 6. Summary of water quality data from laboratory analysis of water samples. Note: values are the average of 2 replicates per site. Values in brackets are from the handheld meters used in the field.

Sample Site	Alkalinity (mg/L CaC03)	Conductivity (цS)	pH (pH units)	Dissolved Copper (mg/L)	Total Copper (mg/L)
Detection Limits	10	1	pH units	0.001	0.001
EF02	10	90.6 (75)	7.3 (6.9)	0.040	0.049
EF04	10	45.0 (46)	7.0	0.002	0.004
EF06	14	53.4 (45)	7.1	< 0.001	0.001
EF13	11	40.6 (37)	7.0	< 0.001	0.001
EF14	15	45.5 (45)	7.0	0.001	0.001
Mean	11	57	7.1		

Note: Analysis of water samples was by North Island Labs, Courtenay, BC.

Data loggers for monitoring water temperature were reinstalled at 4 of the 5 sites in the study area during July through September 2007 (Table 1, Figure 1 for locations). The loggers were set to take a reading every 15 minutes. Data from these loggers have not yet been downloaded and therefore will be presented in next year's report.

5. DISCUSSION

The performance measures for assessing the benefits of the proposed fish flow release include fish abundance, fish condition factor, and habitat continuity. With completion of the 2007 field program, the study now has 3 years of before treatment data. The fish flow release (0.25 m^3/s) commenced in January 2008 and thus field activities to be conducted in the summers of 2008 through 2010 will provide 3 years of post treatment data. Figure 6 shows the fish abundance performance measure, expressed as fish density, during the pre flow release years. Data for 1994 are from Griffith (1996) while the remaining data are from this study. Values shown are geometric means with associated 95% confidence limits. The large error bar around the 1994 mean is due to the small sample size (4 sites). The 2005 and 2006 means exclude sites EF01, EF02, and EF03, as fish appeared to avoid this section of river due to elevated copper concentrations from nearby mine tailings. In 2007, fish did not avoid this area and so these sites were included in the means. The data show pre flow release geometric mean densities ranging from 3.7 to 11.2 FPU for fry, and from 0.7 to 2.3 FPU for 1^+ parr.

As an additional reference point, Figure 6 also shows maximum density estimates for rainbow fry and parr as predicted by the alkalinity model (70 FPU for fry and 11 FPU for parr). Mean densities to date are well below the potential predicted by this model.

The low number of parr in 2007 (0.7 FPU from Figure 6, Chart B) may be due to mortalities incurred from several large spills at Elliott Dam that occurred during the 2006/07 winter, in particular a large flood in November 2006. Figure 7 shows hourly spill flows at Elliott Dam for the winters of 2004/05, 2005/06, and 2006/07 (October 1 to April 30). In 2004/05 and 2005/06 there were only moderate or minor spills (peaks of 215 m^3/s in 2004/05 and 21 m^3/s in 2005/06); however in 2006/07 there were three major spills with peaks ranging from 459 – 636 m^3/s. Although the November 2006 spill was not the largest (peak 459 m^3/s), tributary inflows increased the magnitude of the event in the lower river and BC Hydro operators at the Jordan River Generating Station commented that they had never seen the river that high before (D. Walsh, pers. comm.). Given the magnitude and frequency of flood events during the 2006/07 winter, it seems plausible that mortalities were incurred among overwintering rainbow parr and that this affected parr abundance in summer 2007. Also, 2007 was the first year in which coho fry were not captured at site EF04 and it is possible that their eggs were scoured from the gravel beds during the 2006/07 flood events as well. Thus, the occurrence of large spill events at Elliott Dam may have some influence on the fish abundance performance measure.

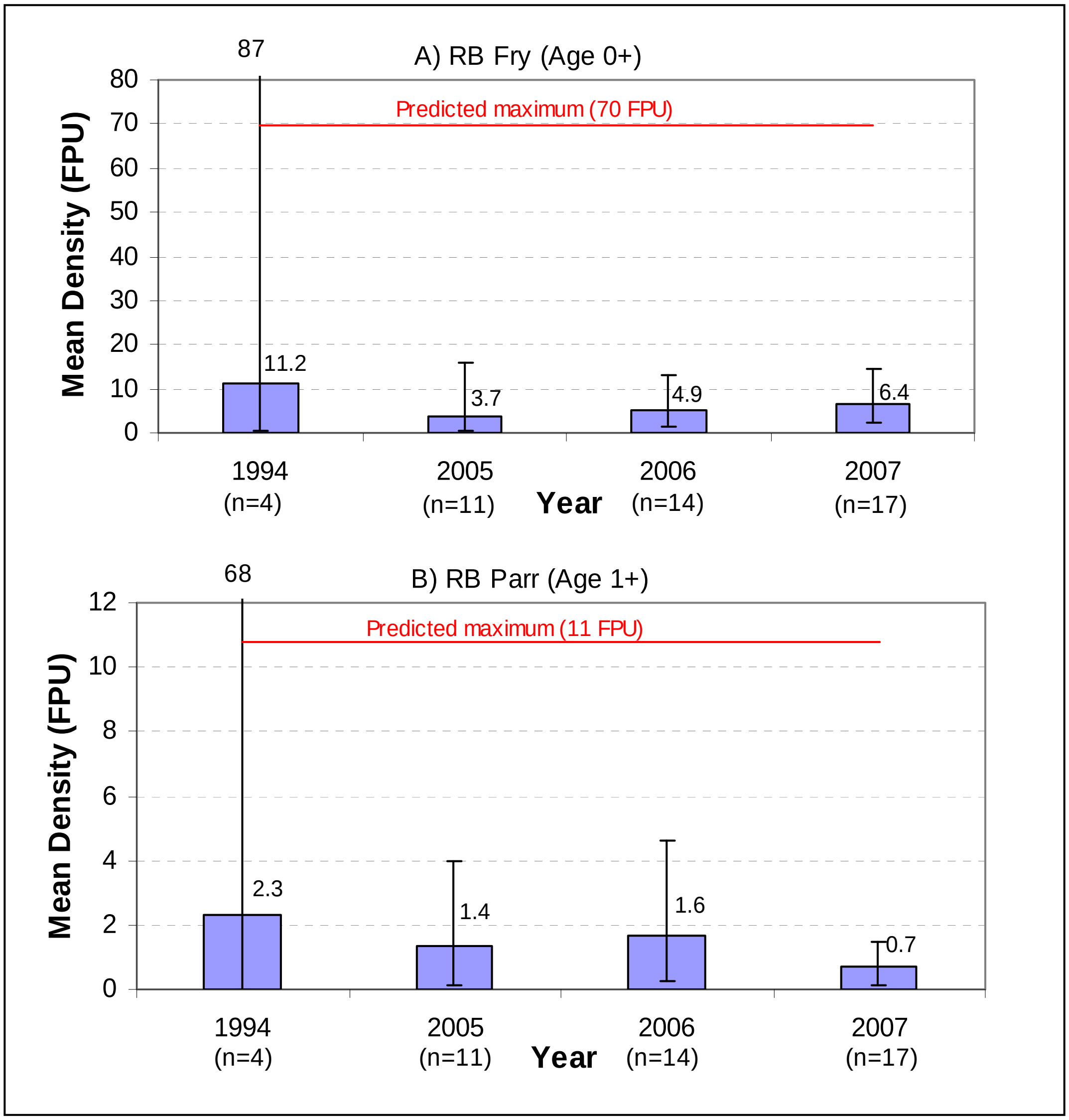

Figure 6. Geometric mean density (fish/100 m^2) of fry (Chart A) and 1$^+$ parr (Chart B) for study years to date. Error bars represent 95% confidence limits; "n" refers to the number of sites per year; FPU refers to fish per 100 m^2.

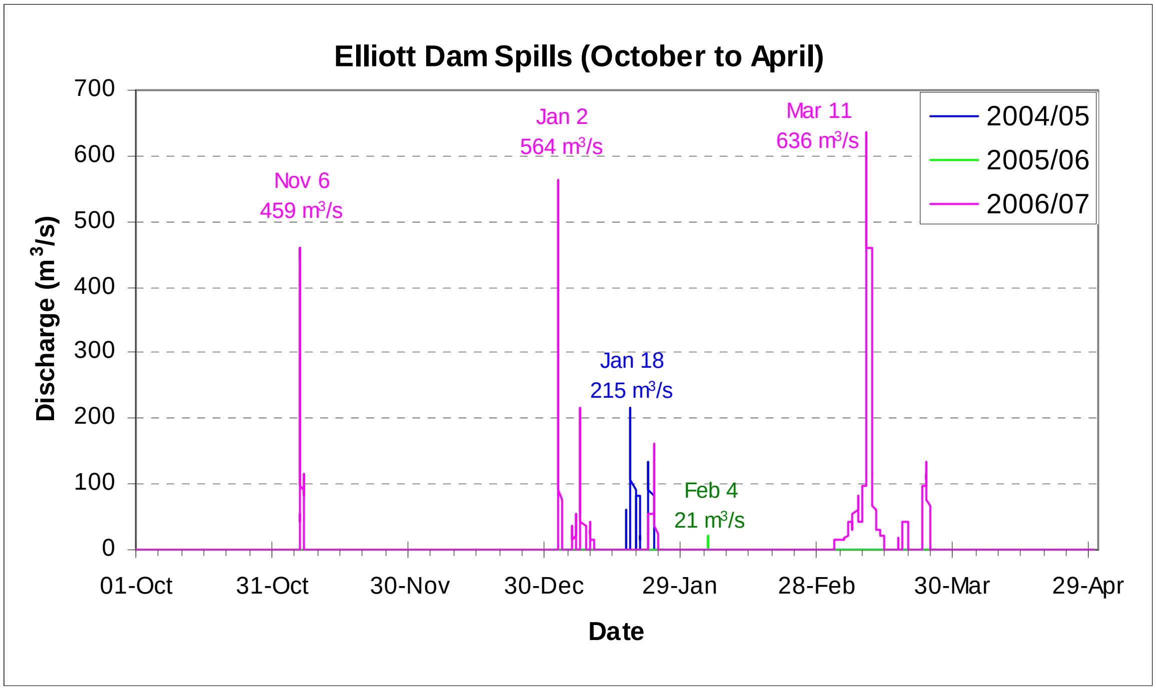

Figure 7. Hourly non-turbine releases (spills) at Elliott Dam during the winters of 2004/05, 2005/06, and 2006/07 (October 1 to April 30 period) (data supplied by BC Hydro).

The second performance measure, fish condition factor, is shown for pre flow release years in Figure 8. Values represent the arithmetic mean of all ages of rainbow trout combined for each year. These data show that rainbow trout condition factor prior to initiation of the fish flow release ranged from 1.05 in 1994 to 1.10 in 2006. Analysis of variance indicated no significant difference among means (α = 0.05). It should be noted that the 2007 condition factor was negatively biased by fish from Sites EL01, EL02, and EL03. Fish captured at these sites exhibited extremely poor condition factor (average 0.90, n = 21), probably due to elevated copper concentrations as previously discussed. If these 21 fish are excluded from the analysis, the overall mean condition factor for 2007 is 1.11, which is very close to the value found in 2006.

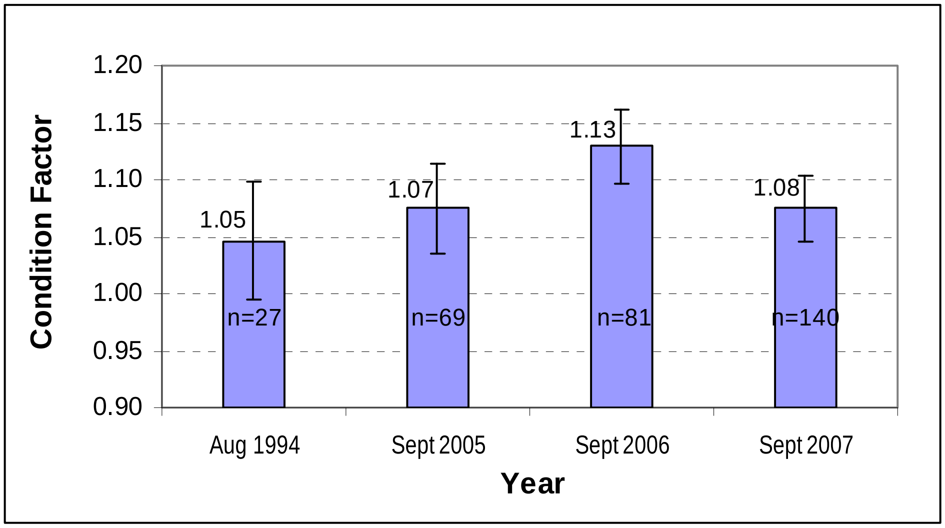

Figure 8. Mean condition factor of rainbow trout for study years to date. Error bars indicate 95% confidence limits; "n" refers to the sample size.

With respect to the third performance measure, habitat continuity, the habitat surveys estimated that 93% of the lower Jordan River stream length was wetted in 2007 (i.e., 7% with subsurface flows) and wetted area amounted to 68,102 m^2 (excluding Reach 4) (Table 4). These values are greater than in 2006 when wetted stream length was 85% (15% subsurface) and wetted area amounted to 54,617 m^2. Differences between years is due to higher flows in 2007 (data from the Reach 5 gauge suggests late August flows were 2.3 fold higher in 2007 compared with 2006).

Connectivity is not solely related to the occurrence of subsurface flows. During the 2005 and 2006 surveys, we noted that the ability of fish to move from one habitat unit to the next was severely limited at summer base flows. This was due to conditions at pool and glide tailouts where flows were often reduced to a trickled through the substrate with insufficient depth or volume for fish passage. We concluded that this condition probably restricted the dispersal of fry and parr until fall rains arrived. In 2007 base flows were about 2.3 fold higher than in 2006, and we noted that the connectivity from one habitat unit to the next was improved.

6. RECOMMENDATIONS

The main recommendation stemming from the 2007 field program is that future habitat surveys should be collected at a consistent flow, ideally at the summer base flow. In 2007, habitat data for Reaches 2 and 5 were collected 10 days after the other reaches and flows were about 38% higher at that time. This complicates the ability to relate habitat statistics to a given flow regime.

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Appendix A. Catch statistics, population estimates, and associated density and biomass calculations for electrofishing sites completed in September 2007.

Site No.	Reach	Date	Time	Habitat	Area(m^2)	Species	Age	No. Passes	Pass1	Pass2	Pass3	Pass4	N	MnWt	1-Pass Exp	PopEst	PopVar	PopSE	PopCV	PopLO	PopUP	CapEst	CapVar	CapSE	CapCV	CapLO	CapUP
															Population Statistics							Capture Statistics					
EF01	R2	08/09/2007	10:00	Glide/Riff	70.38	RB	0	1	3	-	-	-	3	2.0	1.262	4	0.00	0.06		3.00	6.08						
EF02	R2	08/09/2007	11:30	Glide	111.70	RB	0	1	1	-	-	-	1	1.1	1.262	1	0.00	0.02		1.00	3.07						
EF03	R2	08/09/2007	17:00	Glide	60.39	RB	0	1	17	-	-	-	17	1.5	1.262	21	0.12	0.35		18.80	23.20						
EF04	R3	08/09/2007	13:15	Glide	90.00	RB	0	3	37	6	4	-	47	1.9		47	0.81	0.90	0.02	47.00	48.76	0.77	0.00	0.08	0.06	0.64	0.90
EF05B	R4	07/09/2007	15:15	Glide	61.30	RB	0	3	3	1	0	-	4	4.2		4	0.04	0.21	0.05	4.00	4.40	0.80	0.04	0.26	0.21	0.40	1.20
EF05C	R4	07/09/2007	13:30	Glide	81.00	RB	0	3	2	0	0	-	2	6.7		2	0.00	0.00	0.00	2.00	2.00	1.00	0.00	0.00	0.00	1.00	1.00
EF05D	R5	07/09/2007	11:00	Glide	76.00	RB	0	3	5	1	0	-	6	2.2		6	0.02	0.14	0.02	6.00	6.28	0.86	0.02	0.17	0.14	0.58	1.14
EF06	R5	06/09/2007	16:30	Glide/Riff	34.00	RB	0	3	12	3	2	-	17	3.3		17	0.73	0.85	0.05	17.00	18.67	0.71	0.01	0.17	0.12	0.47	0.95
EF07	R5	06/09/2007	15:21	Glide/Riff	38.97	RB	0	1	2	-	-	-	2	6.4	1.262	3	0.00	0.04		2.00	5.08						
EF08	R5	06/09/2007	14:00	Glide/Riff	62.43	RB	0	1	5	-	-	-	5	5.8	1.262	6	0.01	0.10		5.00	8.09						
EF09	R5	06/09/2007	11:00	Glide/Riff	54.72	RB	0	3	4	1	0	-	5	3.5		5	0.03	0.17	0.03	5.00	5.33	0.83	0.03	0.20	0.17	0.50	1.16
EF11	R6	05/09/2007	13:00	Glide	71.15	RB	0	3	6	3	1	-	10	2.4		10	0.74	0.86	0.09	10.00	11.68	0.67	0.03	0.26	0.17	0.33	1.00
EF12	R6	05/09/2007	15:30	Glide	41.08	RB	0	3	4	0	1	-	5	1.9		5	0.20	0.44	0.09	5.00	5.87	0.71	0.05	0.31	0.22	0.28	1.15
EF13	R6	05/09/2007	17:15	Glide	54.00	RB	0	1	4	-	-	-	4	2.9	1.262	5	0.01	0.08		4.00	7.08						
EF14B	R6	04/09/2007	15:00	Glide	80.01	RB	0	1	0	-	-	-	0	-		0											
EF14	R6	04/09/2007	13:45	Glide	81.94	RB	0	1	0	-	-	-	0	-		0											
EF15	R7	04/09/2007	12:00	Pool	48.38	RB	0	3	0	0	0	-	0	-		0											
Arithmetic Mean														2.5													EF01 - EF15:
Geometric Mean																											EF01 - EF15:
EF01	R2	08/09/2007	10:00	Glide/Riff	70.38	RB	1	1	0	-	-	-	0	-		0											
EF02	R2	08/09/2007	11:30	Glide	111.70	RB	1	1	0	-	-	-	0	-		0											
EF03	R2	08/09/2007	17:00	Glide	60.39	RB	1	1	0	-	-	-	0	-		0											
EF04	R2	08/09/2007	13:15	Glide	90.00	RB	1	3	4	0	0	0	4	23.0		4	0.00	0.00	0.00	4.00	4.00	1.00	0.00	0.00	0.00	1.00	1.00
EF05B	R4	07/09/2007	15:15	Glide	61.30	RB	1	3	0	0	0	-	0	-		0											
EF05C	R4	07/09/2007	13:30	Glide	81.00	RB	1	3	0	0	0	-	0	-		0											
EF05D	R5	07/09/2007	11:00	Glide	76.00	RB	1	3	0	0	0	-	0	-		0											
EF06	R5	06/09/2007	16:30	Glide/Riff	34.00	RB	1	3	2	0	0	-	2	25.9		2	0.00	0.00	0.00	2.00	2.00	1.00	0.00	0.00	0.00	1.00	1.00
EF07	R5	06/09/2007	15:21	Glide/Riff	38.97	RB	1	1	0	-	-	-	0	-	1.262	0											
EF08	R5	06/09/2007	14:00	Glide/Riff	62.43	RB	1	1	3	-	-	-	3	21.4	1.262	4	0.00	0.06		3.00	6.08						
EF09	R5	06/09/2007	11:00	Glide/Riff	54.72	RB	1	3	1	0	0	-	1	16.3		1	0.00	0.00	0.00	1.00	1.00	1.00	0.00	0.00	0.00	1.00	1.00
EF11	R6	05/09/2007	13:00	Glide	71.15	RB	1	3	1	0	0	-	1	14.5		1	0.00	0.00	0.00	1.00	1.00	1.00	0.00	0.00	0.00	1.00	1.00
EF12	R6	05/09/2007	15:30	Glide	41.08	RB	1	3	1	0	0	-	1	30.7		1	0.00	0.00	0.00	1.00	1.00	1.00	0.00	0.00	0.00	1.00	1.00
EF13	R6	05/09/2007	17:15	Glide	54.00	RB	1	1	0	-	-	-	0	-	1.262	0											
EF14B	R6	04/09/2007	15:00	Glide	80.01	RB	1	1	0	-	-	-	0	-	1.262	0											
EF14	R6	04/09/2007	13:45	Glide	81.94	RB	1	1	0	-	-	-	0	-	1.262	0											
EF15*	R7	04/09/2007	12:00	Pool	48.38	RB	1	3	0	0	0	-	0	-		0											
Arithmetic Mean														22.4													EF4 - EF15:
Geometric Mean																											EF4 - EF15:

Site No.	FPU	FPU_{LO}	FPU_{UP}	BPU	BPU_{LO}	BPU_{UP}	PUW	AFPU	$AFPU_{LO}$	$AFPU_{UP}$	ABPU	$ABPU_{LO}$	$ABPU_{UP}$	MxBPU	%Sat
	Observed Density			Observed Biomass			Adjusted Density				Adjusted Biomass				
EF01	5.68	4.26	8.64	11.5	8.7	17.5	38.56%	14.7	11.1	22.4	29.8	22.5	45.5	189	16
EF02	0.90	0.90	2.75	1.0	1.0	3.0	63.21%	1.4	1.4	4.4	1.5	1.5	4.8	189	1
EF03	34.77	31.13	38.41	51.5	46.1	56.9	40.32%	86.2	77.2	95.3	127.6	114.3	141.0	189	68
EF04	52.22	52.22	54.18	98.7	98.7	102.4	31.70%	164.7	164.7	170.9	311.3	311.3	323.0	189	165
EF05B	6.53	6.53	7.18	27.4	27.4	30.2	10.06%	64.9	64.9	71.4	272.6	272.6	299.9	189	144
EF05C	2.47	2.47	2.47	16.4	16.4	16.4	34.66%	7.1	7.1	7.1	47.2	47.2	47.2	189	25
EF05D	7.89	7.89	8.26	17.6	17.6	18.4	10.11%	78.1	78.1	81.7	174.2	174.2	182.2	189	92
EF06	50.00	50.00	54.91	164.0	164.0	180.1	46.30%	108.0	108.0	118.6	354.2	354.2	389.0	249	142
EF07	7.70	5.13	13.03	49.3	32.8	83.4	51.21%	15.0	10.0	25.4	96.0	64.0	162.6	249	39
EF08	9.61	8.01	12.95	55.9	46.6	75.4	50.83%	18.9	15.8	25.5	110.0	92.0	148.4	249	44
EF09	9.14	9.14	9.74	31.8	31.8	33.9	39.21%	23.3	23.3	24.8	81.1	81.1	86.3	249	33
EF11	14.05	14.05	16.42	34.2	34.2	39.9	15.00%	93.7	93.7	109.5	227.7	227.7	266.1	294	77
EF12	12.17	12.17	14.29	22.9	22.9	26.9	78.56%	15.5	15.5	18.2	29.1	29.1	34.2	294	10
EF13	9.26	7.41	13.11	26.4	21.1	37.4	64.92%	14.3	11.4	20.2	40.8	32.5	57.6	294	14
EF14B	0.00	0.00	0.00	0.0	0.0	0.0	14.85%	0.0	0.0	0.0	0.0	0.0	0.0	294	0
EF14	0.00	0.00	0.00	0.0	0.0	0.0	9.53%	0.0	0.0	0.0	0.0	0.0	0.0	294	0
EF15	0.00	0.00	0.00	0.0	0.0	0.0	9.81%	0.0	0.0	0.0	0.0	0.0	0.0	294	0
Arithmetic Mean	13.08			35.8			35.81%								
Geometric Mean	6.36			15.4											
EF01	0.00	0.00	0.00	0.0	0.0	0.0	17.70%	0.0	0.0	0.0	0.0	0.0	0.0	189	0
EF02	0.00	0.00	0.00	0.0	0.0	0.0	14.20%	0.0	0.0	0.0	0.0	0.0	0.0	189	0
EF03	0.00	0.00	0.00	0.0	0.0	0.0	17.62%	0.0	0.0	0.0	0.0	0.0	0.0	189	0
EF04	4.44	4.44	4.44	102.0	102.0	102.0	8.06%	55.1	55.1	55.1	1264.5	1264.5	1264.5	189	669
EF05B	0.00	0.00	0.00	0.0	0.0	0.0	2.53%	0.0	0.0	0.0	0.0	0.0	0.0	189	0
EF05C	0.00	0.00	0.00	0.0	0.0	0.0	11.76%	0.0	0.0	0.0	0.0	0.0	0.0	189	0
EF05D	0.00	0.00	0.00	0.0	0.0	0.0	0.00%	#DIV/0!	#DIV/0!	#DIV/0!	0.0	0.0	0.0	189	0
EF06	5.88	5.88	5.88	152.1	152.1	152.1	11.50%	51.2	51.2	51.2	1323.5	1323.5	1323.5	249	532
EF07	0.00	0.00	0.00	0.0	0.0	0.0	19.23%	0.0	0.0	0.0	0.0	0.0	0.0	249	0
EF08	6.41	4.81	9.74	136.9	102.7	208.1	16.16%	39.6	29.7	60.3	846.3	634.7	1288.6	249	340
EF09	1.83	1.83	1.83	29.8	29.8	29.8	5.58%	32.8	32.8	32.8	534.6	534.6	534.6	249	215
EF11	1.41	1.41	1.41	20.4	20.4	20.4	2.85%	49.3	49.3	49.3	714.9	714.9	714.9	294	243
EF12	2.43	2.43	2.43	74.7	74.7	74.7	36.41%	6.7	6.7	6.7	205.7	205.7	205.7	294	70
EF13	0.00	0.00	0.00	0.0	0.0	0.0	13.44%	0.0	0.0	0.0	0.0	0.0	0.0	294	0
EF14B	0.00	0.00	0.00	0.0	0.0	0.0	1.03%	0.0	0.0	0.0	0.0	0.0	0.0	294	0
EF14	0.00	0.00	0.00	0.0	0.0	0.0	0.00%	#DIV/0!	#DIV/0!	#DIV/0!	0.0	0.0	0.0	294	0
EF15*	0.00	0.00	0.00	0.0	0.0	0.0	0.00%	#DIV/0!	#DIV/0!	#DIV/0!	0.0	0.0	0.0	294	0
Arithmetic Mean	1.32			30.3			10.47%								
Geometric Mean	0.68			3.5											

[Back side of 11 x 17 appendix]

Appendix B. Electrofishing site summaries.

APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF01

General Information

Date	08-Sep-07	Stream:	Jordan River	
Time	10:00	Reach:	R2	
Crew	DB,JJ	UTM:	422,236	5,364,601

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.21
Sample Width (m):	4.63	Mean Velocity (m/s):	0.070
Sample Length (m):	15.2	Metered Discharge (cms):	0.062
Sample Area (m2):	70.38	Gauge Station Discharge (cms):	0.0450
Temperature (C):	12.8	Stage (% MAD):	0.3
Conductivity (microS):	74		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	4.7
Gradient (%):	2.0%	Channel Width (m):	32.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
35	0	0	0	0	0	100	0

Substrate (% each category):

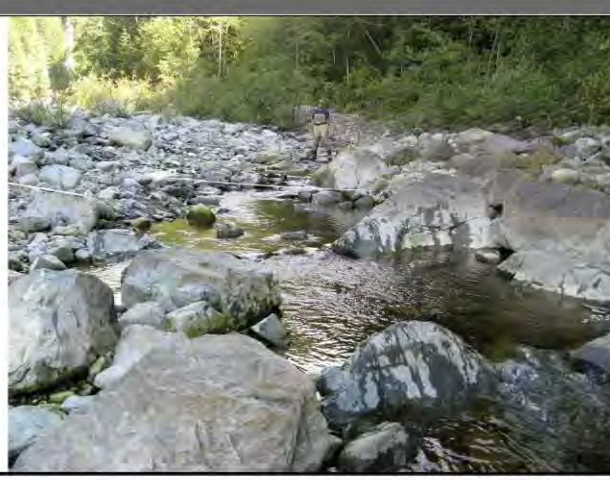
Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
10	25	15	30	20	0	110	160

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	61	2.0	3				3	4.00	5.68	4.26	8.64	11.56	0.39

Photo

EF01



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF02

General Information

Date	08-Sep-07	Stream:	Jordan River
Time	11:30	Reach:	R2
Crew	DB,JJ	UTM:	422,236 5,364,629

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.09
Sample Width (m):	5.88	Mean Velocity (m/s):	0.111
Sample Length (m):	19.0	Metered Discharge (cms):	0.046
Sample Area (m2):	111.70	Gauge Station Discharge (cms):	0.0450
Temperature (C):	13.1	Stage (% MAD):	0.3
Conductivity (microS):	75		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	5.88
Gradient (%):	2.0%	Channel Width (m):	44.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
30	0	0	0	0	33	67	0

Substrate (% each category):

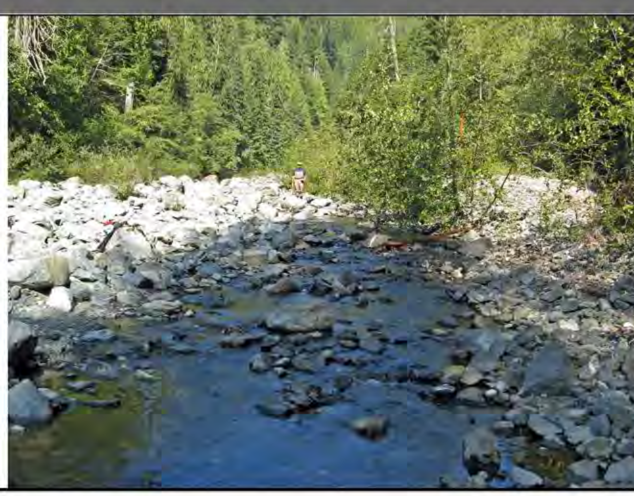
Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
15	15	30	30	10	0	50	90

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	53	1.1	1				1	1.00	0.90	0.90	2.75	0.98	0.63
CC	99	53		1				1	1.00	0.90	0.90	2.75		

Photo

EF02



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF03

General Information

Date	08-Sep-07	Stream:	Jordan River
Time	17:00	Reach:	R2
Crew	DB,JJ	UTM:	422,236 5,364,690

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.28
Sample Width (m):	6.10	Mean Velocity (m/s):	0.039
Sample Length (m):	9.9	Metered Discharge (cms):	0.059
Sample Area (m2):	60.39	Gauge Station Discharge (cms):	0.0450
Temperature (C):	14.3	Stage (% MAD):	0.3
Conductivity (microS):	80		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	6.1
Gradient (%):	2.0%	Channel Width (m):	40.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
20	0	0	0	0	0	100	0

Substrate (% each category):

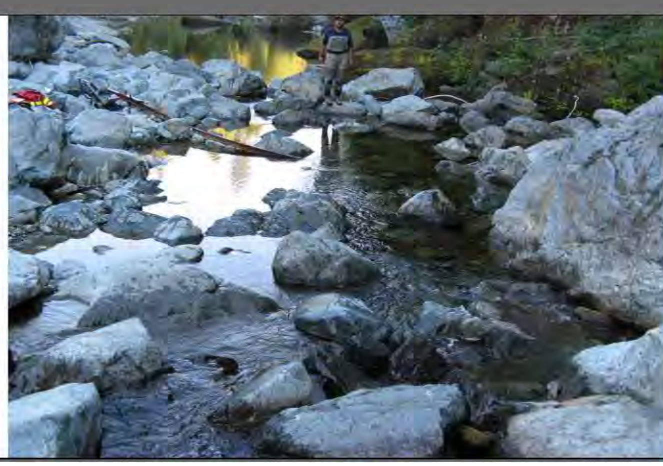
Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
15	25	10	20	30	0	180	470

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	53	1.5	17				17	21.00	34.77	31.13	38.42	51.55	0.40

Photo

EF03



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF04

General Information

Date	08-Sep-07	Stream:	Jordan River
Time	13:15	Reach:	R2
Crew	DB,JJ	UTM:	422,188 5,364,930

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.26
Sample Width (m):	6.00	Mean Velocity (m/s):	0.021
Sample Length (m):	15.0	Metered Discharge (cms):	0.044
Sample Area (m2):	90.00	Gauge Station Discharge (cms):	0.0440
Temperature (C):	15.4	Stage (% MAD):	0.3
Conductivity (microS):	46		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	7
Gradient (%):	0.2%	Channel Width (m):	36.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
35	0	0	0	0	0	100	0

Substrate (% each category):

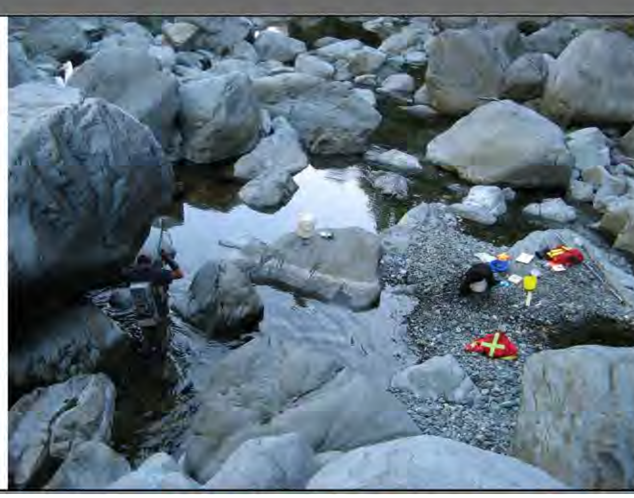
Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
15	20	35	15	15	0	390	440

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	56	1.9	37	6	4		47	47.00	52.22	52.22	54.18	98.89	0.32
RB	1	130	23.0	4	0	0		4	4.00	4.44	4.44	4.44	102.00	0.08
CC	99	85		0	2	0		2	2.00	2.22	2.22	4.48		

Photo

EF04



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF05B

General Information

Date	07-Sep-07	Stream:	Jordan River
Time	15:15	Reach:	R5
Crew	DB,JJ	UTM:	422,418 5,365,805

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.47
Sample Width (m):	6.13	Mean Velocity (m/s):	0.009
Sample Length (m):	10.0	Metered Discharge (cms):	0.023
Sample Area (m2):	61.30	Gauge Station Discharge (cms):	0.0590
Temperature (C):	14.8	Stage (% MAD):	
Conductivity (microS):	44		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	6.13
Gradient (%):	0.0%	Channel Width (m):	3.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
25	0	0	0	0	0	100	0

Substrate (% each category):

Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
15	30	20	20	15	0	200	250

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	69	4.2	3	1	0		4	4.00	6.53	6.53	7.18	27.41	0.10

Photo

EF05B

APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF05C

General Information

Date	07-Sep-07	Stream:	Jordan River
Time	13:30	Reach:	R4
Crew	DB,JJ	UTM:	422,447 5,365,860

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.31
Sample Width (m):	5.40	Mean Velocity (m/s):	0.031
Sample Length (m):	15.0	Metered Discharge (cms):	0.046
Sample Area (m2):	81.00	Gauge Station Discharge (cms):	0.0590
Temperature (C):	13.7	Stage (% MAD):	
Conductivity (microS):	37		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	5.4
Gradient (%):	0.0%	Channel Width (m):	25.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
50	0	0	0	0	0	100	0

Substrate (% each category):

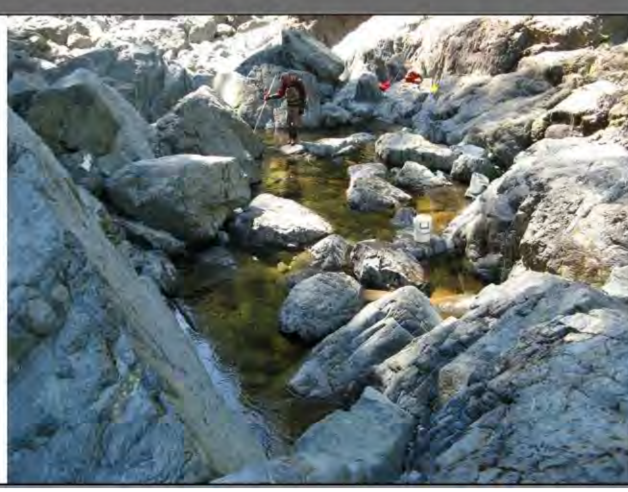
Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
5	10	10	10	60	5	290	420

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	80	6.7	2	0	0		2	2.00	2.47	2.47	2.47	16.42	0.35

Photo

EF05C



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF05D

General Information

Date	07-Sep-07	Stream:	Jordan River
Time	11:00	Reach:	R4
Crew	DB,JJ	UTM:	422,470 5,365,885

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.40
Sample Width (m):	6.08	Mean Velocity (m/s):	0.000
Sample Length (m):	12.5	Metered Discharge (cms):	0.000
Sample Area (m2):	76.00	Gauge Station Discharge (cms):	
Temperature (C):	14.4	Stage (% MAD):	
Conductivity (microS):	45		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	7
Gradient (%):	0.0%	Channel Width (m):	29.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
11	0	0	0	0	9	91	0

Substrate (% each category):

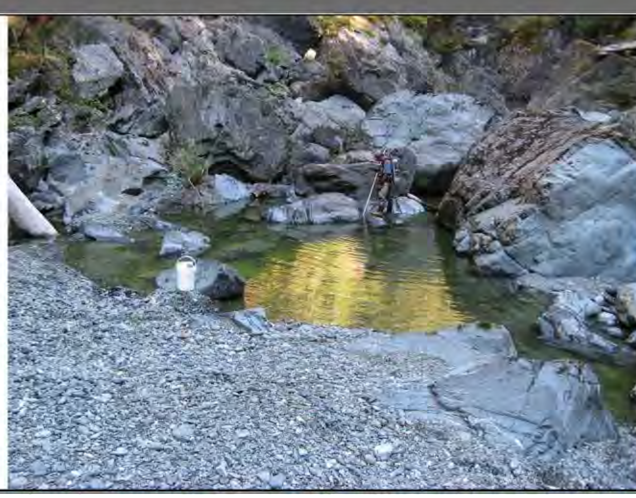
Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
15	35	25	10	12	3	100	500

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	59	2.2	5	1	0		6	6.00	7.89	7.89	8.26	17.63	0.10

Photo

EF05D



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF06

General Information

Date	06-Sep-07	Stream:	Jordan River
Time	16:30	Reach:	R5
Crew	DB,JJ	UTM:	423,650 5,366,900

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.22
Sample Width (m):	3.40	Mean Velocity (m/s):	0.029
Sample Length (m):	10.0	Metered Discharge (cms):	0.017
Sample Area (m2):	34.00	Gauge Station Discharge (cms):	0.0340
Temperature (C):	15.9	Stage (% MAD):	
Conductivity (microS):	45		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	3.4
Gradient (%):	0.1%	Channel Width (m):	26.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
60	0	0	0	0	0	100	0

Substrate (% each category):

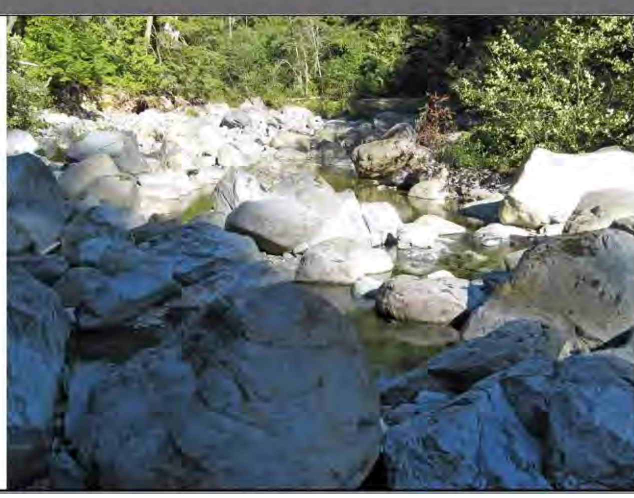
Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
5	10	10	15	60	0	200	290

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	65	3.3	12	3	2		17	17.00	50.00	50.00	54.91	163.82	0.46
RB	1	123	25.9	2	0	0		2	2.00	5.88	5.88	5.88	152.06	0.12

Photo

EF06



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF07

General Information

Date	06-Sep-07	Stream:	Jordan River
Time	15:21	Reach:	R5
Crew	DB,JJ	UTM:	423,790 5,366,954

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.17
Sample Width (m):	4.33	Mean Velocity (m/s):	0.087
Sample Length (m):	9.0	Metered Discharge (cms):	0.071
Sample Area (m2):	38.97	Gauge Station Discharge (cms):	0.0360
Temperature (C):	17.5	Stage (% MAD):	
Conductivity (microS):	47		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	4.3
Gradient (%):	0.5%	Channel Width (m):	27.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
60	0	0	0	0	0	100	0

Substrate (% each category):

Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
0	5	5	40	50	0	160	195

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	79	6.4	2				2	3.00	7.70	5.13	13.04	49.27	0.51

Photo

EF07

APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF08

General Information

Date	06-Sep-07	Stream:	Jordan River	
Time	14:00	Reach:	R5	
Crew	DB,JJ	UTM:	423,843	5,367,050

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.12
Sample Width (m):	3.83	Mean Velocity (m/s):	0.082
Sample Length (m):	16.3	Metered Discharge (cms):	0.042
Sample Area (m2):	62.43	Gauge Station Discharge (cms):	0.0360
Temperature (C):	16.4	Stage (% MAD):	
Conductivity (microS):	44		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	4
Gradient (%):	0.1%	Channel Width (m):	30.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
61	0	0	0	0	2	98	0

Substrate (% each category):

Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
5	20	10	20	45	0	150	160

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	78	5.8	5				5	6.00	9.61	8.01	12.96	55.93	0.51
RB	1	125	21.4	3				3	4.00	6.41	4.81	9.74	136.90	0.16

Photo

EF08

APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF09

General Information

Date	06-Sep-07	Stream:	Jordan River
Time	11:00	Reach:	R5
Crew	DB,JJ	UTM:	423,924 5,367,218

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.10
Sample Width (m):	3.80	Mean Velocity (m/s):	0.026
Sample Length (m):	14.4	Metered Discharge (cms):	0.014
Sample Area (m2):	54.72	Gauge Station Discharge (cms):	0.0400
Temperature (C):	14.7	Stage (% MAD):	
Conductivity (microS):	39		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	4.5
Gradient (%):	0.0%	Channel Width (m):	33.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
27	0	0	0	0	7	93	0

Substrate (% each category):

Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
10	25	15	25	25	0	96	150

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	65	3.5	4	1	0		5	5.00	9.14	9.14	9.74	31.80	0.39
RB	1	117	16.3	1	0	0		1	1.00	1.83	1.83	1.83	29.79	0.06

Photo

EF09

APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF11

General Information

Date	05-Sep-07	Stream:		Jordan River
Time	13:00	Reach:		R6
Crew	DB,JJ	UTM:	425,101	5,367,855

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.37
Sample Width (m):	5.27	Mean Velocity (m/s):	0.006
Sample Length (m):	13.5	Metered Discharge (cms):	0.015
Sample Area (m2):	71.15	Gauge Station Discharge (cms):	0.0296
Temperature (C):	16.2	Stage (% MAD):	
Conductivity (microS):	40		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	5.3
Gradient (%):	0.5%	Channel Width (m):	25.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
60	0	0	0	0	0	100	0

Substrate (% each category):

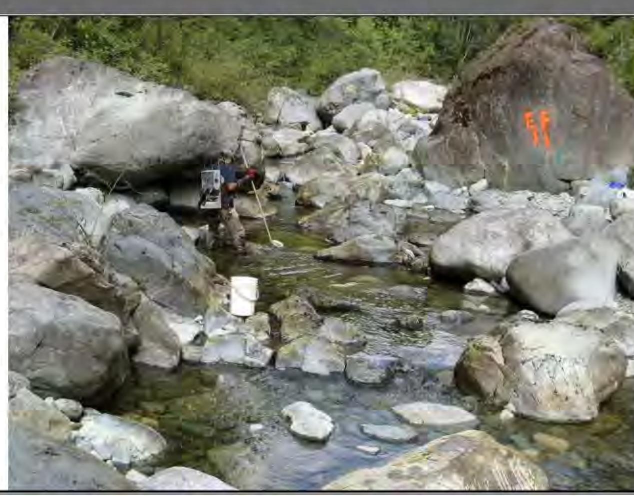
Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
0	5	15	15	60	5	160	400

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	59	2.4	6	3	1		10	10.00	14.05	14.05	16.42	34.15	0.15
RB	1	108	14.5	1	0	0		1	1.00	1.41	1.41	1.41	20.38	0.03

Photo

EF11



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF12

General Information

Date	05-Sep-07	Stream:	Jordan River
Time	15:30	Reach:	R6
Crew	DB,JJ	UTM:	425,155 5,367,982

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.25
Sample Width (m):	4.37	Mean Velocity (m/s):	0.095
Sample Length (m):	9.4	Metered Discharge (cms):	0.080
Sample Area (m2):	41.08	Gauge Station Discharge (cms):	0.0296
Temperature (C):	15.9	Stage (% MAD):	
Conductivity (microS):	19		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	4.4
Gradient (%):	3.0%	Channel Width (m):	37.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
60	0	0	0	0	0	100	0

Substrate (% each category):

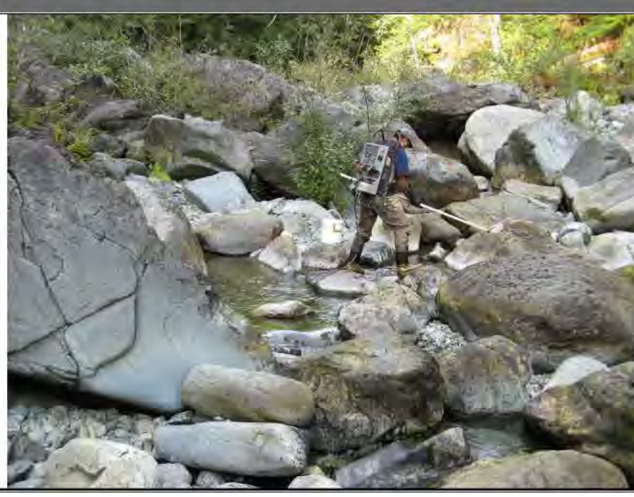
Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
0	0	15	25	60	0	113	220

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	56	1.9	4	0	1		5	5.00	12.17	12.17	14.29	22.88	0.79
RB	1	140	30.7	1	0	0		1	1.00	2.43	2.43	2.43	74.73	0.36

Photo

EF12



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF13

General Information

Date	05-Sep-07	Stream:	Jordan River
Time	17:15	Reach:	R6
Crew	DB,JJ	UTM:	425,292 5,368,220

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.11
Sample Width (m):	4.03	Mean Velocity (m/s):	0.074
Sample Length (m):	13.4	Metered Discharge (cms):	0.044
Sample Area (m2):	54.00	Gauge Station Discharge (cms):	0.0296
Temperature (C):	15.9	Stage (% MAD):	
Conductivity (microS):	37		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	8
Gradient (%):	0.0%	Channel Width (m):	27.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
25	0	0	0	0	0	100	0

Substrate (% each category):

Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
15	35	10	10	25	5	130	200

Fish Population Characteristics

Species	Age	MnLgth	MnWt	C1	C2	C3	C4	N	Pop	FPU	LCL_FPU	UCL_FPU	BPU	PUW
RB	0	63	2.9	4				4	5.00	9.26	7.41	13.11	26.39	0.65

Photo

EF13

APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF14

General Information

Date	04-Sep-07	Stream:	Jordan River	
Time	13:45	Reach:	R6	
Crew	DB,JJ	UTM:	426,032	5,369,290

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.43
Sample Width (m):	6.30	Mean Velocity (m/s):	0.000
Sample Length (m):	13.6	Metered Discharge (cms):	0.000
Sample Area (m2):	81.94	Gauge Station Discharge (cms):	0.0040
Temperature (C):	13.7	Stage (% MAD):	
Conductivity (microS):	45		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	
Gradient (%):	0.0%	Channel Width (m):	29.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
35	0	0	0	0	0	100	0

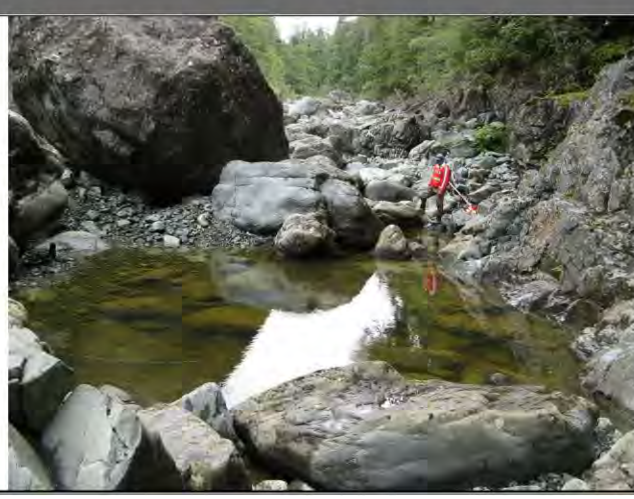
Substrate (% each category):

Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
0	5	30	10	25	30	165	230

Fish Population Characteristics

Photo

EF14



APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF14B

General Information

Date	04-Sep-07	Stream:	Jordan River	
Time	15:00	Reach:	R6	
Crew	DB,JJ	UTM:	425,872	5,369,082

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.33
Sample Width (m):	6.00	Mean Velocity (m/s):	0.002
Sample Length (m):	12.7	Metered Discharge (cms):	0.003
Sample Area (m2):	80.01	Gauge Station Discharge (cms):	0.0034
Temperature (C):	15.2	Stage (% MAD):	
Conductivity (microS):	41		

Meso Habitat Characteristics

Habitat Type:	G	Stream Width (m):	6.3
Gradient (%):	1.0%	Channel Width (m):	31.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
45	0	0	0	0	0	100	0

Substrate (% each category):

Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
0	5	10	25	60	0	220	250

Fish Population Characteristics

Photo

EF14B

APPENDIX B: ELECTROFISHING SITE SUMMARY

Site: EF15

General Information

Date	04-Sep-07	Stream:		Jordan River
Time	12:00	Reach:		R7
Crew	DB,JJ	UTM:	426,114	5,369,525

Sample Site Characteristics

MS or SC:	MS	Mean Depth (m):	0.48
Sample Width (m):	4.10	Mean Velocity (m/s):	0.000
Sample Length (m):	11.8	Metered Discharge (cms):	0.000
Sample Area (m2):	48.38	Gauge Station Discharge (cms):	0.0042
Temperature (C):	14.8	Stage (% MAD):	
Conductivity (microS):	57		

Meso Habitat Characteristics

Habitat Type:	P	Stream Width (m):	0
Gradient (%):	0.0%	Channel Width (m):	31.0

Cover (% Total Cover and % each category):

Total Cover	Deep Pool	LWD	SWD	Instream Veg.	Overstream Veg.	Boulders	Cut Banks
25	0	0	0	0	0	100	0

Substrate (% each category):

Fines	Small Gravel	Large Gravel	Cobble	Boulders	Bedrock	D90 (cm)	Dmax (cm)
0	5	10	10	15	60	240	300

Fish Population Characteristics

Photo

EF15